

Bernard K. Appiah

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PROFESSIONAL SUMMARY

Emerging biostatistician with three (3) years of experience in statistical modeling, public-health data analysis, and quantitative research. Skilled in survival methods, machine learning, and regression, with master's-level training in advanced modeling, statistical inference, and high-dimensional data analysis. Background includes hands-on experience in teaching, statistical consulting, and field data collection, with a proven ability to communicate technical results to non-statistical audiences and to collaborate effectively across disciplines and cultures. Driven to pursue advanced research in statistical methodology and to develop methods that address real-world scientific and public health problems.

EDUCATION

- University of Minnesota Duluth (UMD)** **Duluth, MN**
MS in Statistics; GPA: 4.0/4.0 **August 2025 - Present**
 - Research: Posterior Variable Selection for ODE-Based Survival Models via Dynamic Hazard Preservation
 - Advisor: Prof. Dananjani. M. Liyanage
- University of Ghana** **Accra, Ghana**
B.sc Mathematics and Statistics (Double Major); GPA: 3.64/4.00 **September 2019 - September 2023**
 - Research: Application of Machine Learning Models to Identify Malaria Risk Factors Among Pregnant Women in Ghana
 - Advisor: Dr. Gabriel Kallah-Dagadu
 - First Class Honors

TEACHING EXPERIENCE

- Department of Mathematics & Statistics, University of Minnesota Duluth** **Duluth, MN**
Teaching Assistant **August 2025– Present**
 - Designed and delivered 225+ lectures for Calculus I (MATH 1296), Calculus II (MATH 1297), Calculus III (MATH 1298), and Differential Equations (MATH 3280), reaching 300+ undergraduates across 3 semesters – Fall 2025, Spring 2026, and Fall 2026.
 - Offered one-on-one support during office hours, working closely with 20–35 students per section to tackle challenging problems and build confidence in the subject matter.
 - Provided detailed, constructive feedback on student assignments and exams, helping students identify areas for improvement while celebrating their progress.
 - Conducted targeted review sessions and lab sessions using Mathematica and Python, reinforcing key concepts and strengthening students' computational skills.
- Department of Actuarial Science and Statistics, University of Ghana** **Accra, Ghana**
Teaching Assistant & Research Assistant **November 2023 – December August 2024**
 - Led instruction in statistical methods I & II (STAT 333 & STAT 334), and regression analysis (STAT 445) for 400+ undergraduates who found inference and regression the hardest content in the major, covering both lecture and discussion sections.

- Provided 10+ hours weekly of one-on-one R and SPSS guidance for students unable to translate coursework theory into working code, serving as the department's main applied-computing support channel.
- Supported faculty research spanning multiple concurrent projects by gathering and cleaning raw datasets that arrived incomplete and inconsistently formatted.
- Ran analyses in R and SPSS on those cleaned datasets, producing the summaries and model output that fed directly into faculty research objectives.

Ghana Baseline Evaluation Survey, University of Ghana
Enumerator and Data Analyst

Accra, Ghana
January 2024 - February 2024

- Conducted door-to-door surveys to collect data on the livelihood of respondents in various communities in the Eastern region.
- Analyzed the survey results using R software to help make informed decisions.
- Maintained detailed records of survey progress, ensuring accurate and complete documentation.

RESEARCH EXPERIENCE

Manuscripts Under Review

- **Appiah, B. K.**, Incoom-Koomson, K., Bonney, S., Boye, M., & Kallah-Dagadu, G. (2026). Application of Machine Learning Models to Identify the Risk Factors of the Incidence of Malaria Among Pregnant Women in Ghana ([Link](#)).

Graduate Thesis (in progress)

Posterior Variable Selection for ODE-Based Survival Models via Dynamic Hazard Preservation

- Developing ODE-based survival models using damped harmonic oscillators to characterize dynamic hazard functions.
- Incorporating covariate effects and implementing maximum likelihood and Bayesian estimation.
- Developing posterior directional variable selection methods to identify influential covariates and their effect directions.
- Conducting Monte Carlo simulations to evaluate estimation accuracy, selection performance, and directional error control under right censoring.

Undergraduate Research

Application of Machine Learning Models to Identify the Risk Factors of the Incidence of Malaria Among Pregnant Women in Ghana ([Link](#))

- Applied Random Forest, Decision Trees, KNN, Naive Bayes, and Neural Networks to 2019 Ghana Malaria Indicator Survey data to identify risk factors for malaria incidence among pregnant women.

Academic Projects

- *Tossing Accuracy Among Three Participants* ([Link](#))
 - Analyzed recorded toss distances using one-way ANOVA, regression, and residual diagnostics.
 - Compared participants with Bonferroni, Tukey, and Scheffé confidence intervals.

- *Comparative Statistical and Machine Learning Approaches for Life Expectancy Prediction in Africa*. Modeled determinants across 54 countries (832 country-year observations); benchmarked via stratified train-test splits, repeated 10-fold cross-validation, and bootstrapping, with Random Forest and Gradient Boosting outperforming traditional regression ([Link](#)).
- *Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators*. Applied multiple linear regression to identify adult mortality and HIV incidence as significant predictors of life expectancy across African countries. ([Link](#)).
- *A Project Proposal to Investigate the Social, Economic, and Legal Aspects of the Galamsey Problem in Ghana*. Proposed a research design covering environmental degradation, socioeconomic impact on mining communities, and the regulatory framework.

Presentations

- *Application of Machine Learning Models to Identify Malaria Risk Factors Among Pregnant Women in Ghana*. Poster Presentation, Swenson College of Science and Engineering (SCSE) Student Research Day, University of Minnesota Duluth. February 2026.
- *Modeling Life Expectancy in Africa Using Socioeconomic and Health Indicators*. Class Presentation, University of Minnesota Duluth. December 2025.

Talks

- *Application of Machine Learning Models to Identify Malaria Risk Factors Among Pregnant Women in Ghana*. Departmental Seminar, University of Ghana, Accra, Ghana. November 2023.
- *Understanding the Least Squares Method using Linear Algebra*. Class Presentation, University of Minnesota Duluth. April 2026.
- *Cramér–Rao Lower Bound and Efficiency of Estimators*. Class Presentation, University of Minnesota Duluth. September 2025.
- *A Statistical Look at Barcelona's Performance: Last Season vs This Season*. Class Presentation, University of Minnesota Duluth. August 2025.

AWARDS AND HONORS

- 2026 – Student Research Grant, University of Minnesota Duluth
- 2026 – Fostering Diversity in Biostatistics Workshop Funding, ENAR
- 2023 – Winner, Mathematics Society Quiz Competition, University of Ghana
- 2023 – Winner, Statistics and Actuarial Science Quiz Competition

ACADEMIC SERVICE, LEADERSHIP, & VOLUNTEERISM

- 2021 – Member, Academic Committee, University of Ghana Actuarial and Statistics Society
- 2020 – Member, The Mathematics Society, Department of Mathematics, University of Ghana

NON-DEGREE EDUCATION & CERTIFICATIONS

- Exploratory Data Analysis in R – DataCamp
- Introduction to Regression in R – DataCamp
- Survival Analysis in R – DataCamp

HIGHLIGHTED SKILLS

- **Soft Skills:** Collaboration, Leadership, Analytical Thinking, and Effective Communication
- **Technical Skills:** Statistical Modeling (R, Python, SPSS), $\LaTeX$, GitHub and Microsoft 365

PROFESSIONAL MEMBER ASSOCIATION

- American Statistical Association
- American Mathematical Society
- Institute of Mathematical Statistics
- Eastern North American Region of the International Biometric Society